GERONTOLOGY & HOME NURSING

GERONTOLOGY

- **Definition:**
- This is a branch of medicine that studies the diseases and other health concerns of the older people.
- It can also be defined as scientific study of problems of ageing from birth to death in all aspects.

Conti……

- Or ageing process which begins with conception and ends with death.
- It is a branch of science that deals with ageing and the problems of ageing people.
- It can also be defined as a branch of medicine devoted to the study of medical problems and care of older people.
- It encompasses the medical aspects of older people.

GERIATRICS

- **Definition:**
- A branch of medicine that deals with prevention, treatment of diseases and disabilities of older people.

When does a person become old?

- The traditional designation for old age is 65 years without any basis in biology.
- Age 65 was chosen as retirement age in Europe but 60 in Ghana.
- The question as to when a person becomes old can be answered in different ways.

Old Age

- Chronological age is based solely on a person's age in years. This help to predict many health problems.
- Biological age refers to changes in the body that commonly occurs as people age.
- Psychological age is based on how people act and feel.

Why does the body changes?

- The body changes with aging because changes occurs in individual cells and in whole organs.
- These changes results in changes in function, in appearance, and thus in the experience of ageing.

Ageing Cells

- As cells age, their function decline. Eventually, they must die, as part of the body's functioning.

Ageing organs

- How well organs function depends on how well the cells within them function.
- Older cells function less well.
- In some organs, cells die and are not replaced, therefore the number of cells decreases.
- For instance the cells in the testes, ovaries, liver, and kidneys decreases markedly in as the body ages.

Why does the function decline?

- For most people most of the time, the decline in organ function does not affect the body's ability to function during normal daily activities.
- People usually notice the decline in organ function only when very demanding tasks are attempted or when a disorder develops.

How the Body Ages

- Other changes are hardly noticeable. E.g. the kidneys may find it difficult to filter blood well as they do previously.

THEORIES OF AGEING

- Theories make sense of phenomena; they give order and perspectives from which to view facts.
- Although all gerontological practitioners draw from many of the same theories, each discipline has a unique approach to assessing situations, planning care, and solving problems.
- The three main theories of ageing are biological, sociological and psychological theories (Miller 1999).

Biological Theories

- Biological theories of ageing address questions about the basic ageing processes that affect all living organisms.
- They look at various natural happenings of the body such as diseases, stress, nutrition and environmental factors as contribution to ageing.
- This is relevant today, looking at eating patterns, obesity which can shorten life and also exposes one to diseases of their heart, kidneys and liver.

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- Some of the biological theories include:
- Wear and tear theory
- Genetic theory
- Cross-linkage theory
- Autoimmune theory
- Single organ theory
- The free-radicals theory
- Longevity and senescence theories.

Wear and Tear Theory

- This is a mechanistic theory, the wear and tear theory suggests that organisms have fixed amount of energy available, and that they will wear out on a scheduled basis.
- When enough cells wear out, the body does not function well.
- The wearing-out process is aggravated by harmful stress factors such as smoking, poor diet, muscular strain or excessive intake of alcohol.

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- According to this theory, the body can be likened to machine that is expected to function well when it is new, but that will wear-out with time.
- Parts can be fixed or replaced, but eventually, the machine no longer functions because of extensive accumulation of wear and tear.
- Like the machine, the longevity of the human body will be affected by the care it receives, as well as, by its genetic components.

Theories Based on Genetics

- Some genetic theories of ageing indicate that life expectancy is genetically pre-programmed within specific range of its kind.
- In humans, for instance, the programme allows for a maximum of about 110 years.
- Hayflick, as stated in Miller (1999) estimated that, normal cells divide 50 times in this number of years.
- Moreover, cells are genetically programmed to stop dividing after achieving 50 cell division, at which time they begin to deteriorate.

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- According to him, abnormal cells, however are not subject to this predictable programme and can divide an indefinite number of times.
- Some genetic theories called mutations theories suggest that ageing is a result of mutation of somatic cells or alterations in DNA repair mechanisms.

Cross Linkage Theories

- The cross linkage theory proposes that molecular structures that are normally separated may be bound together through chemical reactions.
- For instance, collagen on connective tissue becomes distorted and cross-links as an individual ages.
- This distortion reduces the rate at which nutrients are absorbed and waste disposed off, and reduces the efficiency of nerve pathway.

Theories Based on Auto Immune System

- Immunity theories are based on the knowledge that immune system components are affected by the ageing process.
- Diminished immune functioning can lead to an increase in the body’s autoimmune response.

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- Because of the age-related, diminished function of the immune system, the older person has less defense against foreign organisms and thus has increased susceptibility to cancer, infections, rheumatoid arthritis or maturity onset diabetes.
- Autoimmunity theory could explain the fact that older adults often manifest allergies to food and environmental conditions that they previously never experienced.

Single-Organ Theory

- This theory views ageing as a disease related failure of a vital body organ.
- Based on this theory, people do not die of old age, they die because of a disease.
- Or routine wear and tear causes a vital part of the body to cease functioning while the rest of the body is still capable of living.

Free Radical Theory

- The human body has natural protective mechanisms or antioxidants that slow the rate of oxidation or counteract the effects of free radicals.
- The free radical theory postulates that protective mechanisms decrease, or free radical formation increase with advancing age.
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- When free radicals attack molecules, they damage the cell membranes; and ageing is thought to occur because of accumulated cell damage that eventually interferes with function.

Longevity and Senescence Theories

- Several models try to explain longevity and senescence theories.
- Some propose that long life is influenced by factors such as genetic; physical environment; physical activity throughout life; consumption of moderate amount of alcohol; sexual activity persisting in to advanced years; dietary factors and factors relating to social environment.

Conti……

- Additional factors include laughter; low ambitions; belief in God; close family ties, freedom and independence; organised, purposeful behaviour and a positive view of life.
- Another view is that ageing is a universal, progressive, and ultimately fetal disease and 'that illness is a necessary concomitant of being old'.

Conti……

- Illness is to be accepted if one is old.
- According to this model, not only does sickness invariably accompany old age, it is untreatable because of old age.